

brain-drain brain: CM5, two USB 3 hubs, eight bay slots, M.2 bay, power, UI

power-input

12 V DIN input, fuse, TVS, reverse-polarity FET, bulk capacitance (sized for 8 bays)

File: sheets/power-input.kicad\_sch

power-bucks

5V\_SYS and 5V\_HDD (TPS56637), +3V3 and 3V3\_M2 (AP63203), +1V2 LDO

File: sheets/power-bucks.kicad\_sch

cm5

Compute Module 5 (wireless): power, control, GPIO, microSD, USB-C, UART, RTC cell, fan, DIP, LEDs, OLED

File: sheets/cm5.kicad\_sch

usb3-hub-B

USB5744 hub B on CM5 USB3-1, bay slots 5-8

File: sheets/usb3-hub-B.kicad\_sch

bay-slots

Eight bay slots (PCIe x1 sockets, bay-card pinout): USB 3, USB 2, 12 V, 5 V, BAY\_EN, bay LEDs

File: sheets/bay-slots.kicad\_sch

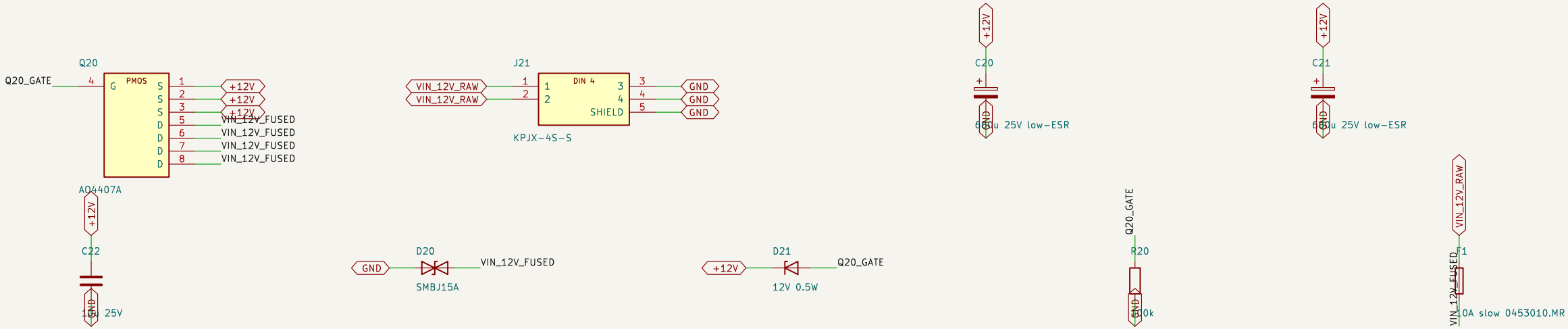
m2-nvme

Bay 9: M.2 M-key on PCIe Gen3 x1, 3V3\_M2 buck + load switch, lid switch

File: sheets/m2-nvme.kicad\_sch

|                                                                                    |         |  |                   |
|------------------------------------------------------------------------------------|---------|--|-------------------|
| brain-drain                                                                        |         |  |                   |
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| Title: brain-drain brain: CM5, two USB 3 hubs, eight bay slots, M.2 bay, power, UI |         |  |                   |
| Size: A3                                                                           | Date:   |  | Rev: v0-generated |
| KiCad E.D.A. 9.0.8                                                                 | Id: 1/8 |  |                   |

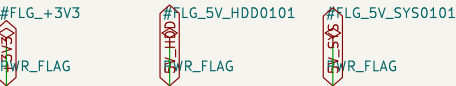
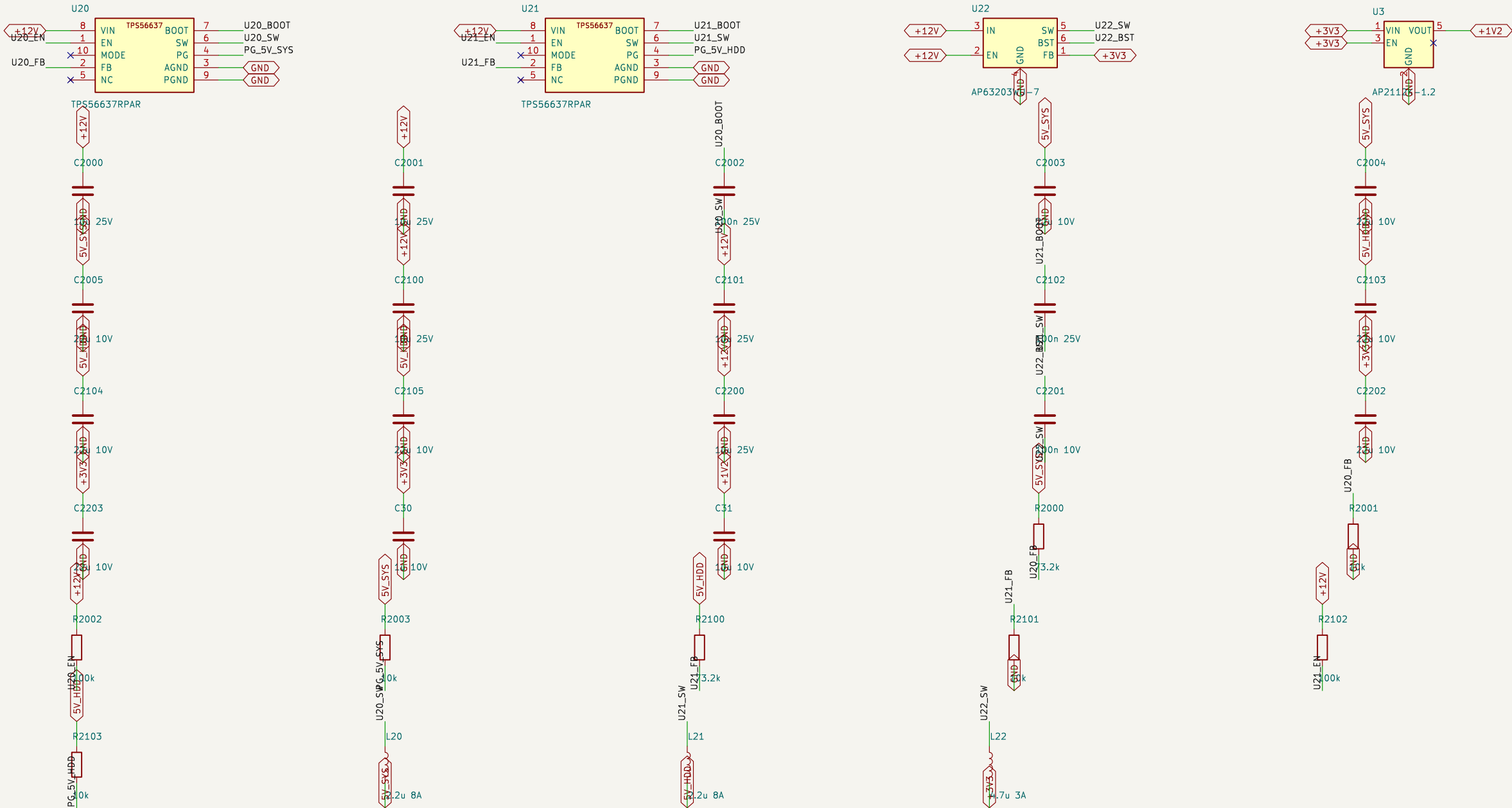
power-input: 12 V DIN input, fuse, TVS, reverse-polarity FET, bulk capacitance (sized for 8 bays)



J21 pins 1/2 = +12 V and 3/4 = GND is a PLACEHOLDER; wire to match the chosen 12 V/10 A brick's DIN pinout (C13).  
Q20 is a P-MOSFET used as an ideal-diode-style reverse-polarity switch; drain to the fused input, source to +12V, gate pulled to GND through R20 and clamped by D21.

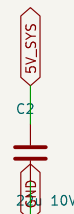
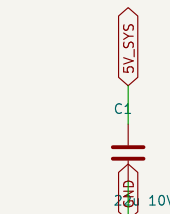
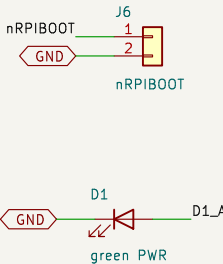
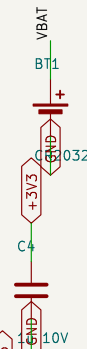
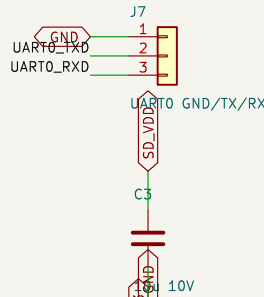
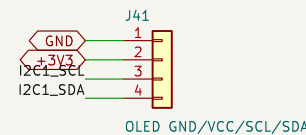
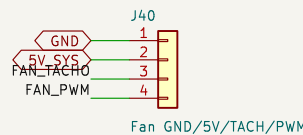
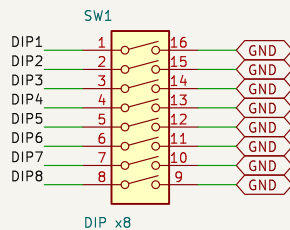
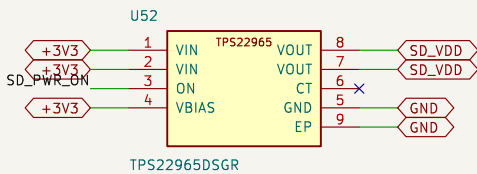
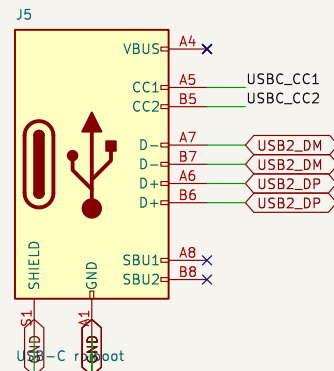
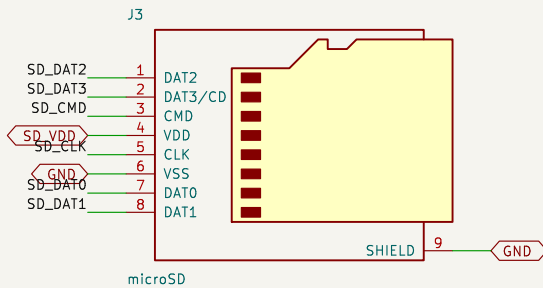
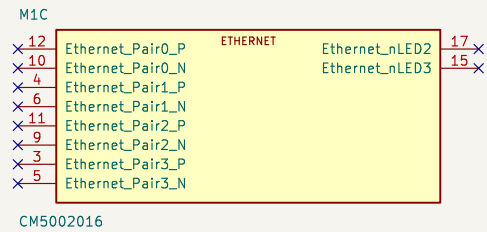
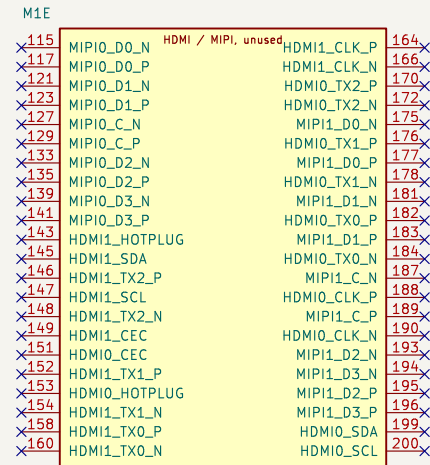
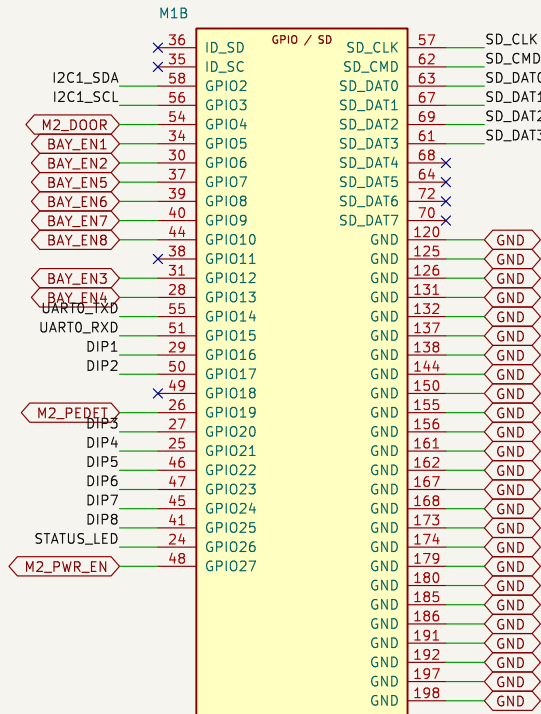
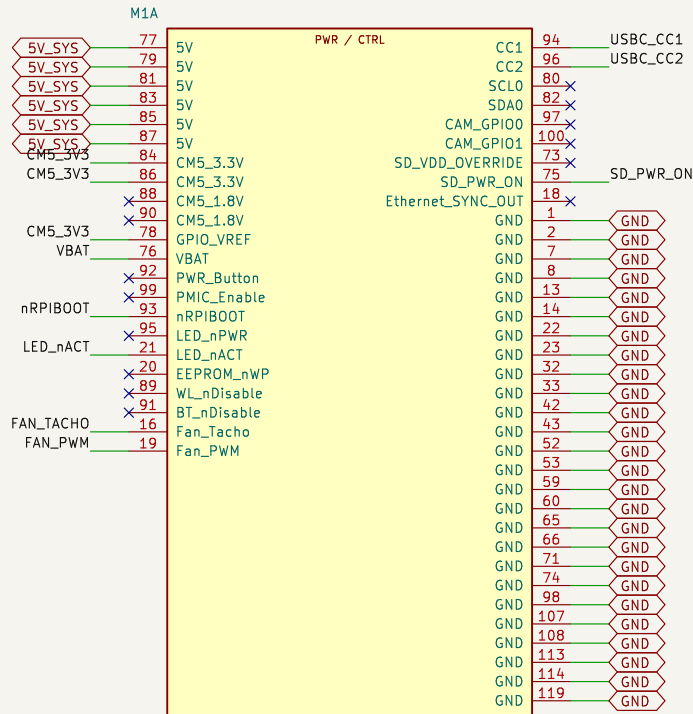
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|-----------------------------------------------------|-------|-------------------|
| brain-drain                                         |       |                   |
| Sheet: /power-input/<br>File: power-input.kicad_sch |       |                   |
| Title: brain-drain - power-input                    |       |                   |
| Size: A3                                            | Date: | Rev: v0-generated |
| KiCad E.D.A. 9.0.8                                  |       | Id: 2/8           |

power-bucks: 5V\_SYS and 5V\_HDD (TPS56637), +3V3 and 3V3\_M2 (AP63203), +1V2 LDO



TPS56637 FB: Vref 0.6 V; 73.2k/10k gives 4.99 V. MODE floating = forced CCM (quietest for the CM5 rail).  
AP63203WU is the fixed 3.3 V variant: FB ties straight to the output. EN tied to VIN (always on).

cm5: Compute Module 5 (wireless): power, control, GPIO, microSD, USB-C, UART, RTC cell, fan, DIP, LEDs, OLED



GPIO\_VREF is tied to the CM5's own 3.3 V output (pins 84/86) for 3.3 V GPIO signalling, per datasheet §4.2.

PMIC\_Enable and PWR\_Button are left floating (internal pull-ups). nRPIBOOT goes to jumper J6.

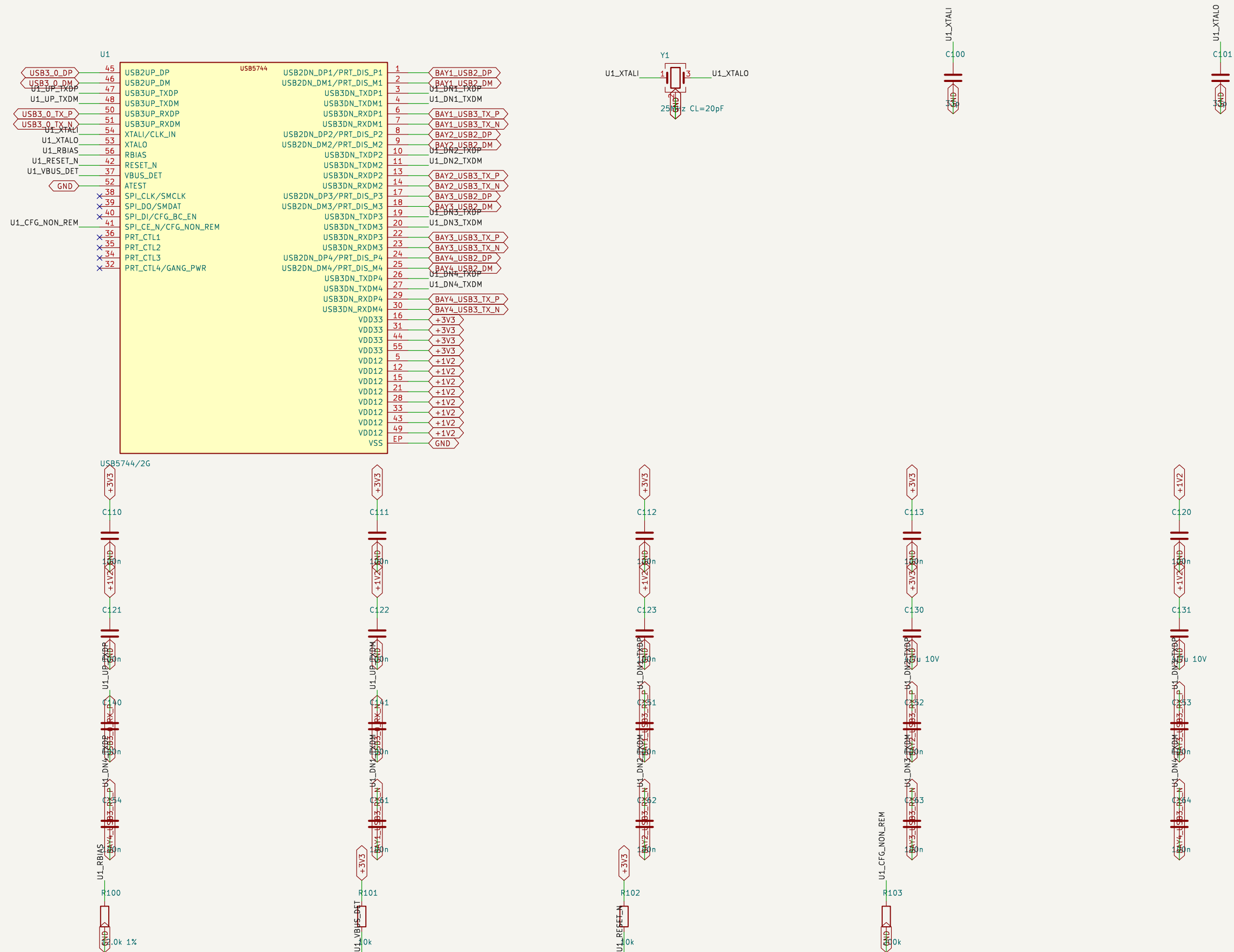
LED\_nACT is an open-drain 20 mA output on the CM5; it sinks D2 directly. GPIO26 sinks D3 (software drives it low to light).

microSD power goes through U52 so the CM5 can power-cycle the card on reboot via SD\_PWR\_ON, matching the CM5IO reference (which uses an RT9742).

Ethernet dropped (C22). The CM5 wireless variant's PCB antenna is on the short module edge that carries MH1; that edge sits on the board's left edge with an 8 mm copper-free strip under it and no metal within 10 mm (CM5 datasheet 4.1.2).

USB-C is a design requirement (not a product). Across the CM5IO, CC1/CC2 connects to itself. VBUS is not connected (the board is powered from 12 V). USB-OTG ID (design requirement)

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usb3-hub-A: USB5744 hub A on CM5 USB3-0, bay slots 1-4
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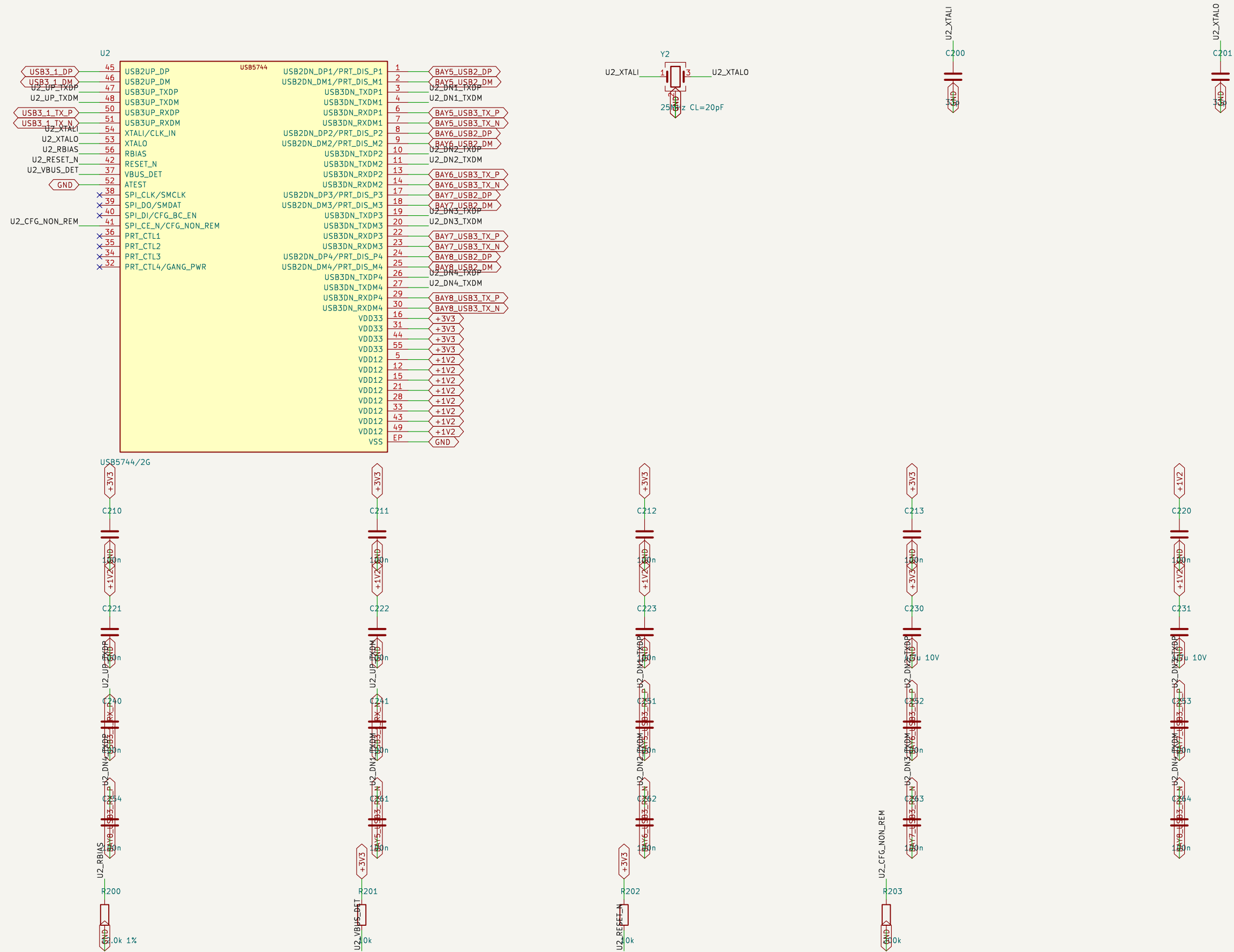


Hub A: upstream on CM5 USB3-0. Downstream ports 1-4 -> bay slots 1-4 (C24: eight slots, four cards populated in v1).

No SPI ROM and no SMBus pull-ups: the hub runs its internal ROM defaults. RESET\_N has a pull-up only; add an RC if strap timing (1 ms hold) is a concern.

Crystal: 25 MHz, CL 20 pF per datasheet table 10-10; 33 pF load caps assume ~3 pF stray, adjust to the chosen crystal.

usb3-hub-B; USB5744 hub B on CM5 USB3-1, bay slots 5-8

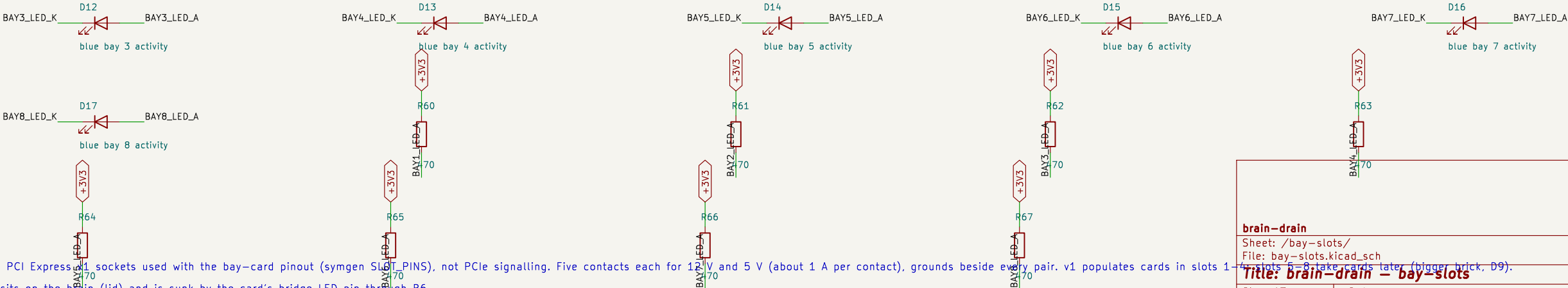
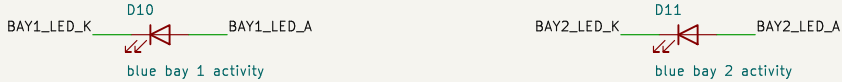
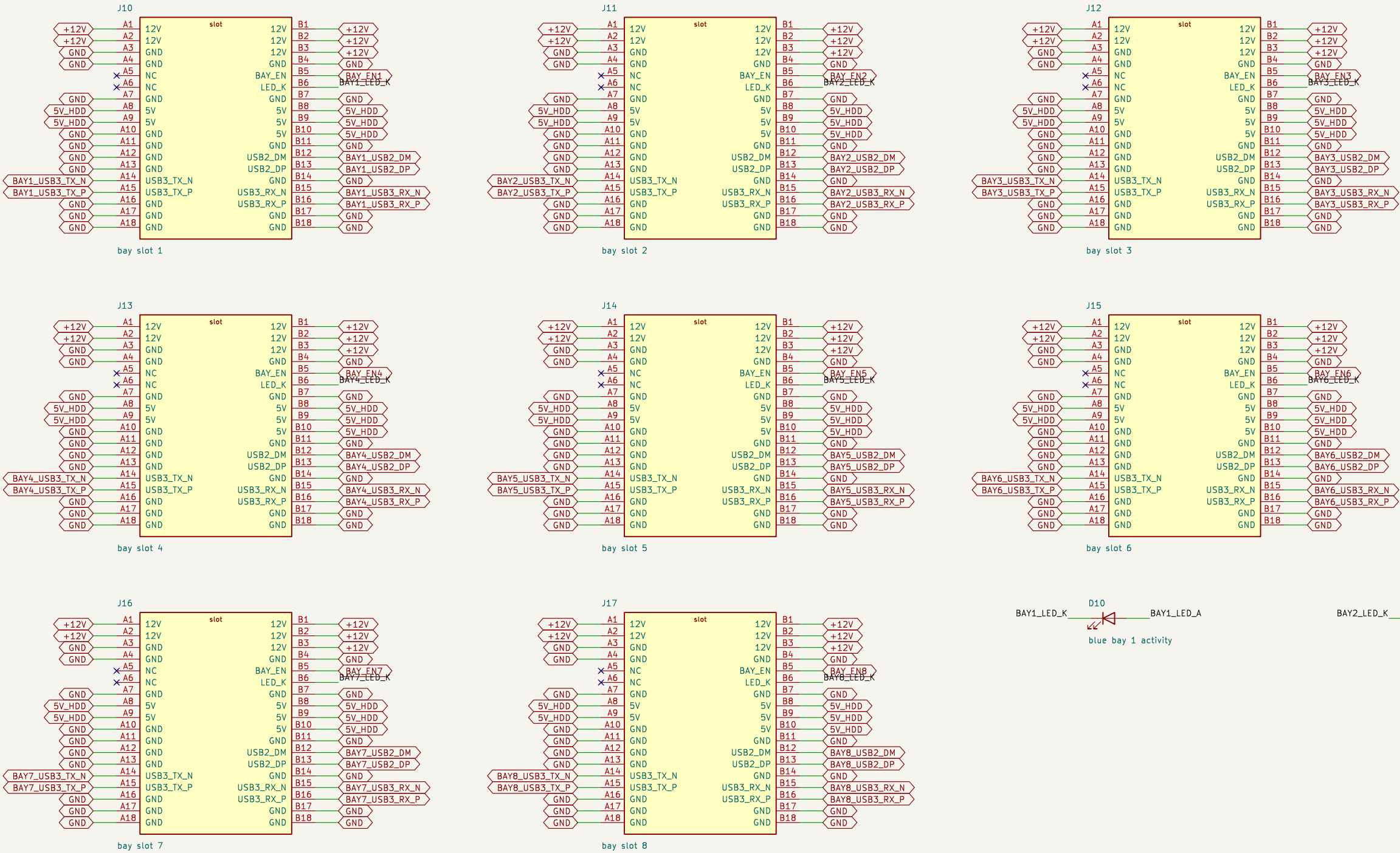


Hub B: upstream on CM5 USB3-1. Downstream ports 1-4 -> bay slots 5-8 (C24: eight slots, four cards populated in v1).

No SPI ROM and no SMBus pull-ups: the hub runs its internal ROM defaults. RESET\_N has a pull-up only; add an RC if strap timing (1 ms hold) is a concern.

Crystal: 25 MHz, CL 20 pF per datasheet table 10-10; 33 pF load caps assume ~3 pF stray, adjust to the chosen crystal.

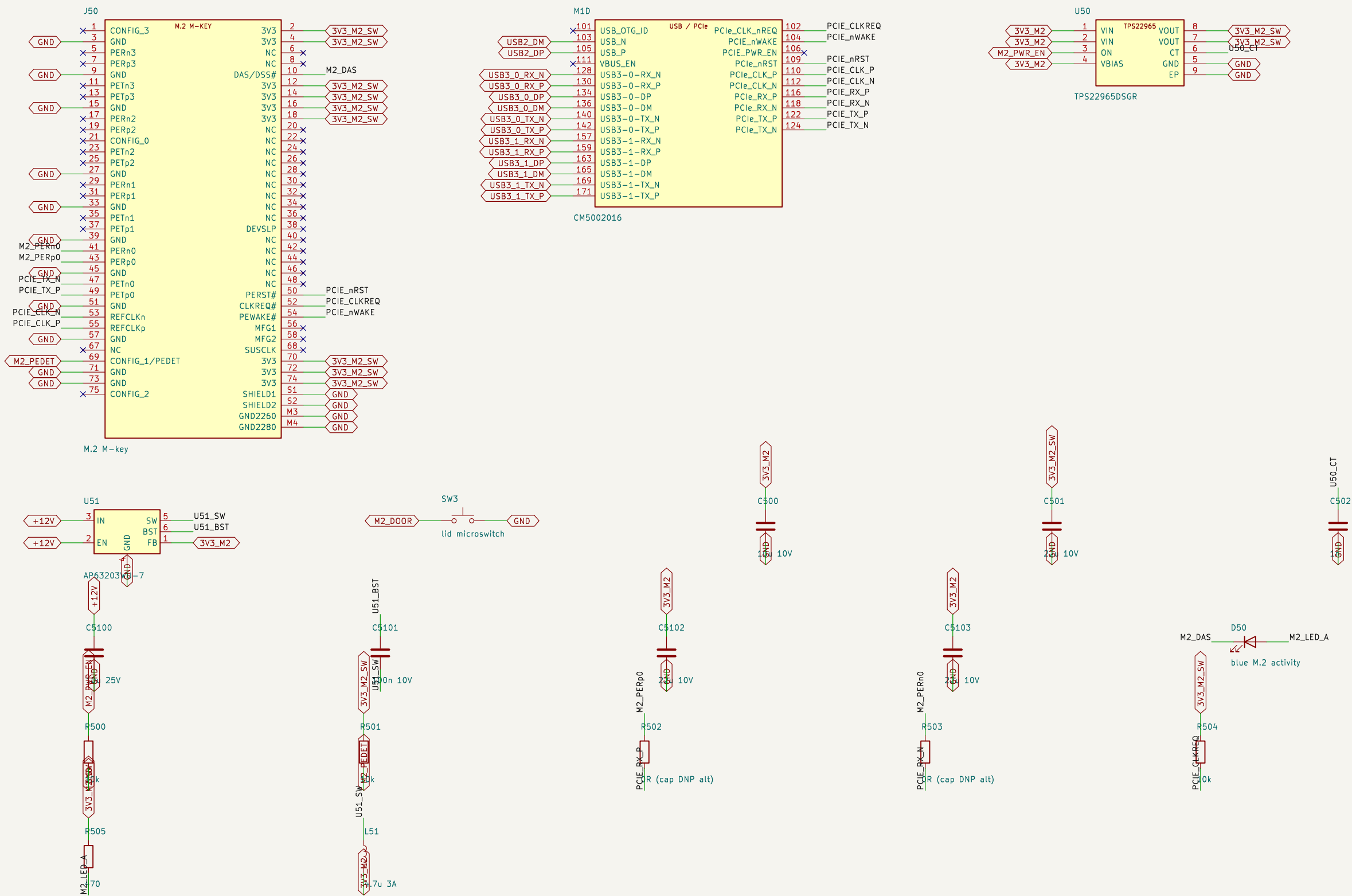
bay-slots: Eight bay slots (PCIe x1 sockets, bay-card pinout): USB 3, USB 2, 12 V, 5 V, BAY\_EN, bay LEDs



Bay slots are PCI Express x1 sockets used with the bay-card pinout (symgen SUGT PINS), not PCIe signalling. Five contacts each for 12 V and 5 V (about 1 A per contact), grounds beside every pair. v1 populates cards in slots 1-4, slots 5-8 take cards later (bigger brick, D9).

The bay LED sits on the Bpin (lid) and is sunk by the card's bridge LED pin through B6.

m2-nvme: Bay 9: M.2 M-key on PCIe Gen3 x1, 3V3\_M2 buck + load switch, lid switch



Polarity: CM5 TX\_P -> PETp0 (49), TX\_N -> PETn0 (47), PERp0 (43) -> RX\_P, PERn0 (41) -> RX\_N. The CM5IO reference wires both pairs inverted (TX\_P to 47, RX\_P to 41), which PCIe link training tolerates; either works.

Only PCIe lane 0 is wired (x1). PET = host transmit (CM5 TX, AC caps on the module). PER = host receive: per the M.2 spec the SSD carries its own TX caps, so R502/R503 are 0R; the same 0402 pads take 100n if a module with its caps turns up.

PEDET (pin 69) is pulled up and read by GPIO19: an M.2 SATA module grounds it, and the software refuses the bay instead of trying PCIe.

Door switch SW3 to GPIO4 (software pull-up): closed = low = door shut. Software powers the slot (M2\_PWR\_EN, GPIO27 -> U50) and rescans PCIe.

brain-drain

Sheet: /m2-nvme/

File: m2-nvme.kicad\_sch

Title: brain-drain - m2-nvme

Author: brain-drain

KiCad E.D.A. 9.0.8

Rev: v0-generated

Id: 8/8